

The Build Log

Nine artifacts, captured from the running system on 13 September 2026

12 COMMITTS, 5 MERGED PRS	138 TESTS PASSING, 0 FAILING	19,541 LINES OF PRODUCT CODE	88.6ms MEDIAN DECISION, 20 RUNS	11.1s FULL SIX-STEP DEMO
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1 Commit history — four phases, one PR each

A sustained build, not one weekend. Scoped to the product commits; the Round 2 submission commit adds documents, not code.

```
$ git log --first-parent --date=short --pretty="%h %ad %s" a13f6cb

a13f6cb 2026-09-12  Fix the three failures that break a deployed demo (#5)
abc1b5c 2026-09-09  Phase 4: demo flow, audit UI, and a real ZK proof (#4)
99bbb96 2026-09-08  Phase 3: autonomous appeals and cryptographic identity (#3)
5f26194 2026-09-07  Phase 2: Insure price transparency (#2)
e48ec54 2026-09-07  Phase 1: autonomous prior authorization core loop (#1)
d5ea3a7 2026-09-07  Initial commit

$ git log --shortstat a13f6cb  # summed

12 commits (6 on main, 6 on phase branches)
170 file changes  +19541 insertions  -367 deletions
```

2 The test suite

138 passing, 0 failing, 23.4 s.

3 Production build

Deployable software — 7 routes, clean compile.

```
$ .venv/bin/python -m pytest

.....
..... [ 52%]
.....
..... [100%]
138 passed, 1 warning in 22.37s
```

```
$ npm run build

✓ Compiled successfully
✓ Generating static pages (8/8)

Route (app)                                Size      First Load JS
└─ ○ /                                       175 B      96.2 kB
└─ ○ /_not-found                           873 B      88.2 kB
└─ ○ /audit                               4.38 kB    91.7 kB
└─ ○ /dashboard                           5.3 kB     101 kB
└─ ○ /demo                                7.62 kB    95 kB
└─ ○ /insure                              5.67 kB    93 kB
└─ f /requests/[id]                        4.14 kB    100 kB
+ First Load JS shared by all              87.4 kB
└─ chunks/117-a45d7ec72ba41ba6.js          31.7 kB
└─ chunks/fd9d1056-2084b62c680baf80.js     53.6 kB
└─ other shared chunks (total)              1.98 kB
f Middleware                                78.6 kB
o (Static)  prerendered as static content
f (Dynamic) server-rendered on demand
```

4 A live authorization decision, end to end

This call: 90.1 ms. Note the patient ID — `mrn-99001` went in, a SHA-256 hash came back.

```
$ curl -s -w " total: %{time_total}s" -X POST localhost:8000/api/auth/request \
-d '{"procedure_code":"27447","diagnosis_code":"M17.11","patient_id":"mrn-99001"}'

{
  "outcome":    "APPROVED",
  "rule_id":    "PAVO-R002",
  "confidence": 1.0,
  "status":     "approved",
  "patient_id": "pat_293e86d31f97652092022a1076aee370"
}
total: 0.090117s
```

Reproduce every figure on this page

`git log --shortstat a13f6cb · cd zk && npm install then cd backend && .venv/bin/python -m pytest · cd frontend && npm run build · the curl shown above.` The ZK tests need `snarkjs` present; without it 7 of the 138 fail on a missing dependency.

A The safety guarantees, demonstrated

5 An unverified message never acts

HTTP 401 before the payload is read.

```
$ # replay an envelope whose payload was
$ # altered after it was signed
$ curl -X POST localhost:8000/api/aria/receive \
  -d @tampered.json

{
  "detail": "Signature does not match the envelope contents."
}
HTTP 401

The rule engine is never reached. The
rejection is written to audit_log as
aria verification failed with the reason.
```

6 Five audit entries per authorization

Every decision row carries its rule ID.

```
$ curl -s localhost:8000/api/audit/$ID

aria_request.created      provider/metro-valley
aria_auth_request.sent    provider/metro-valley
aria_auth_request.received payer/meridian
decision.rendered         payer/meridian
aria_auth_response.sent   payer/meridian

decision.rendered, before -> after:
status:      pending -> approved
rule_id:     None    -> PAV0-R002
confidence:  None    -> 1.0
```

7 A zero-knowledge proof of patient criteria

Three criteria proven. Age, diagnosis and deductible appear nowhere in the response.

```
$ curl -s -X POST localhost:8000/api/zk/generate \
  -d '{"patient_age":42,"diagnosis_code":"M17.11","deductible_met":true}'

{
  "proof_id": "e8e370b4-7dbc-45d3-8b79-abae8961799d",
  "public_signals": [
    "1",
    "1",
    "1",
    "18",
    "196730042053713158391225580434902024231",
    "1"
  ],
  "proof_digest": "sha256:cd2df83a6fe4b231b44e9c223eb88f75747566288720570977ab893ef175eda6",
  "claims": {
    "age_valid": true,
    "diagnosis_valid": true,
    "deductible_valid": true,
    "min_age": "18",
    "deductible_required": true
  },
  "criteria": {
    "min_age": "18",
    "approved_diagnosis_hash": "196730042053713158391225580434902024231",
    "deductible_required": "1",
    "circuit": "patient_criteria",
    "protocol": "groth16"
  },
  "proof": "<groth16 pi_a / pi_b / pi_c - 8 field elements, elided>"
}

NOTE: patient_age, diagnosis_code and deductible_met appear
nowhere in this response. Only the three verdicts do.
```

What this response does and does not reveal

The six public signals are three verdicts (1, 1, 1) and the three policy parameters the payer set (min_age 18, the approved-diagnosis hash, deductible_required 1). The patient's actual age, diagnosis code and deductible balance are never transmitted. The one thing it does leak, stated in §5.3 of the report: because the diagnosis check compares against a single hash, a passing proof tells the payer the patient carries that one diagnosis.

B The running system — one order, start to finish

8 The six-step demo, completed in 11.1 seconds

Every panel is live data. Measured across three consecutive runs: 11.00 s, 11.07 s, 11.17 s.

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DemoDashboardInsureAuditAuth not configured

END-TO-END DEMO

One order, start to finish

Every panel below is live data from the running system — the same agents, rules, signatures, and circuit the rest of the app uses.

Procedure (CPT)

Diagnosis (ICD-10)

27447

M17.11

Run Full Demo

Next

completed in 11.1s

✓ EHR trigger

A physician places an order.

✓ Identity

Both agents prove who they are.

✓ Zero-knowledge

Criteria proven, PHI withheld.

✓ Rule engine

A deterministic rule decides.

✓ Decision

The outcome and its audit trail.

6 Insure

What the patient would actually pay.

STEP 6 — INSURE

Arthroplasty, knee, condyle and plateau (total knee replacement)

CPT 27447 · deductible not met

The same authorization now drives a price comparison for the member — what they would actually pay, at each facility, through insurance or in cash.

#1

Northgate Community Hospital

\$5,296

negotiated \$26,482 · cash \$7,176

Insurance cheaper

#2

Parkside Diagnostic Clinic

\$5,977

negotiated \$29,887 · cash \$11,051

Insurance cheaper

#3

Riverside Imaging & Surgical

\$6,279

negotiated \$31,394 · cash \$11,444

Insurance cheaper

ARIA Protocol v1.0 — every decision carries the rule that produced it.

8a Step 3 — zero-knowledge, in the UI

The same proof as artifact 7, as a payer would see it.

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DemoDashboardInsureAuditAuth not configured

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Diagnosis (ICD-10)

27447

M17.11

Running...

Next

✓ EHR trigger

A physician places an order.

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Both agents prove who they are.

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STEP 3 — ZERO-KNOWLEDGE

What is being proven

The payer needs to know three things about this patient. It does not need the underlying data to know them. The provider sends a mathematical proof instead of the chart.

✓ Patient is at least 18 — the age itself is never sent

✓ Diagnosis matches the covered condition — the code is never sent

✓ Deductible requirement is satisfied — no amount is ever sent

Patient criteria verified without revealing PHI. The payer learns only whether each criterion holds.

The proof

verified

Protocol

groth16 · bn128

Proof

e5dab1d6

Proof digest

sha256:4c69d67fa50c35911b5b4c31bf2865bc3245af4a8cb51ccaedbd30606b100657

Public signals (everything the payer receives)

```
[
  "1",
  "1",
  "1",
  "18",
  "196730042053713158391225580434902024231",
  "1"
]
```

ARIA Protocol v1.0 — every decision carries the rule that produced it.

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C Identity, and the rule that decided

8b Step 2 — identity

Two RSA-2048 signatures, both verified against the sender's registered public key.

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DemoDashboardInsureAuditAuth not configured

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M17.11

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STEP 2 — IDENTITY

Signature verification

verified: true

Each organization holds an RSA-2048 key pair. Pavo stores only the public key — the private key never reaches the database. Every message is signed over a SHA-256 digest of its contents and checked against the sender's registered key before its payload is read.

AUTH_REQUEST

rsa-pss-sha256:UdE0/9b5t30HvcLUS6uKqT0Is1rn8Z0ZY9q6Xow0E6cU998tVLE9qb5AURu9mZMnGsAPokJH0H1X+n6KfIwBxyT61Po+/9Rd1G99XKwj0htdHtEuZAsuZ0XWmuT1Gwa+Rmqn1fvjJ0yp3Gac/rRfe39wpzH2krXBW9CyYQhphWUX2YfIcyTndIendUGDm1QpBkF+29Lc1TkT35bNJPSPFLKteba9MOPHmHd0s4s5aQNDvDxvn3n3kjGhb1UN0Hxte1/QQ/bk0ObTQZ9KgcIN2FqHb0Glb3n5FgwREz03g9uH2z2whVAdrgRRcam221QvMGNU0zeG6Bb6v==

signature verified

AUTH_RESPONSE

rsa-pss-sha256:0ycw0r9JUw6mInYKjVhczpZonn4IFu7SZ/6tB+ZPL/tch70g1uL22ck0LBdsrtP0gfgLh8XJUuFsR8KH/d+rZc583PahfMn+G5vwTlwzvzqf5RBNf878dYshuB25wkv28UcJGXrGSDK30/XGerXPnN4MYpfqsk19ASc31ok81UxryuZ000rDn1Fogc35MhIn2EW0t+zCt9L121S8sW/PW1zkyuj/1HfYAj1bM2CIXb0NSJ37fuyTYa9BMD7hc561LKEp9tZp8S3PY5TK3cJpHmn1KJchQU04VyEuq/UlwadKhtb3v8aM3aISmK3SR6AGJqMnsDs5/5cetK0XVS0==

signature verified

ARIA Protocol v1.0 — every decision carries the rule that produced it.

8c Step 4 — the rule engine

PAVO-R002, with the full text of the rule that produced the approval.

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27447

Diagnosis (ICD-10)

M17.11

Running...

Next

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STEP 4 — RULE ENGINE

Deterministic rule engine

The payer agent evaluates the FHIR bundle against published coverage rules. The first match wins, and the decision carries the rule that produced it.

PAVO-R002

CPT 27447 (total knee arthroplasty) is covered for M17.11 (unilateral primary osteoarthritis, right knee).

Outcome

APPROVED

Confidence

100%

ARIA Protocol v1.0 — every decision carries the rule that produced it.

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D The record behind every decision

9 The dashboard — every decision with the rule that produced it

10 authorizations, 44 ms average resolution, 20 signed ARIA messages.

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Authorization requests

Every decision below records the rule that produced it. Counters refresh every 10 seconds.

Authorizations processed
10
6 auto-approved

Average resolution
44 ms
created to resolved

Appeals win rate
—
none decided yet

ZK proofs generated
5

ARIA messages
20
signed and verified

Submit an authorization request

Stands in for the EHR webhook that fires when a physician places an order.

Procedure code (CPT)
70553

Diagnosis code (ICD-10)
G43.909

Patient ID (hashed, optional)
mrn-12345

Try: MRI brain — approves

Knee replacement, matching Dx — approves

Knee replacement, other Dx — escalates

Office visit — approves

Unknown procedure — escalates

Submit request

Filter: AllPendingApprovedEscalatedDeniedAppealed

STATUS	PROCEDURE	DIAGNOSIS	RULE	CONFIDENCE	SUBMITTED	
Appealed	45378	R51.9	PAVO-R005	100%	9/13/2026, 1:13:05 PM	View trail
Appealed	12345	M17.11	PAVO-R005	100%	9/13/2026, 1:13:05 PM	View trail
Approved	99214	M17.11	PAVO-R004	100%	9/13/2026, 1:13:05 PM	View trail
Approved	99214	Z00.00	PAVO-R004	100%	9/13/2026, 1:13:05 PM	View trail
Appealed	27447	Z00.00	PAVO-R003	100%	9/13/2026, 1:13:05 PM	View trail
Escalated	27447	M17.12	PAVO-R003	100%	9/13/2026, 1:13:05 PM	View trail
Approved	27447	M17.11	PAVO-R002	100%	9/13/2026, 1:13:05 PM	View trail
Approved	70553	Z00.00	PAVO-R001	100%	9/13/2026, 1:13:05 PM	View trail
Approved	70553	R51.9	PAVO-R001	100%	9/13/2026, 1:13:04 PM	View trail
Approved	70553	G43.909	PAVO-R001	100%	9/13/2026, 1:13:04 PM	View trail

Live ARIA feed

Last 10 messages

AUTH_RESPONSE

13:13:05

verified request 8a5b2e89

AUTH_REQUEST

13:13:05

verified request 8a5b2e89

AUTH_RESPONSE

13:13:05

verified request c36762f8

AUTH_REQUEST

13:13:05

verified request c36762f8

AUTH_RESPONSE

13:13:05

verified request 1ccc6da4

AUTH_REQUEST

13:13:05

verified request 1ccc6da4

AUTH_RESPONSE

13:13:05

verified request 3855fb85

AUTH_REQUEST

13:13:05

verified request 3855fb85

AUTH_RESPONSE

13:13:05

verified request fb0a5741

AUTH_REQUEST

13:13:05

verified request fb0a5741

9a The audit trail — 74 events, filterable, exportable

Append-only. Nothing on this screen is editable.

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DemoDashboardInsureAuditAuth not configured

Audit trail

Every event across the system, newest first. Nothing here is editable — the log is append-only.

Export CSV

Entity type

AllAuthorizationsAppealsPrice queriesZK proofsDocuments

From

mm/dd/yyyy

To

mm/dd/yyyy

74 events

9/13/2026, 1:13:11 PM

zk_proof

zk_proof.verified

2da1aa25

agent://pavo/payer/zk-verifier

9/13/2026, 1:13:11 PM

zk_proof

zk_proof.generated

2da1aa25

agent://pavo/provider/zk-prover

9/13/2026, 1:13:10 PM

zk_proof

zk_proof.verified

6977f259

agent://pavo/payer/zk-verifier

9/13/2026, 1:13:10 PM

zk_proof

zk_proof.generated

6977f259

agent://pavo/provider/zk-prover

What to notice in the requests table

Every row carries a rule ID and 100% confidence — 100% because the match is *definitive*, not because a model was sure. No language model participates in any decision on that screen. The one **Escalated** row is PAVO-R003 : a knee replacement whose diagnosis did not match the covered condition, waiting for a human with its record already assembled.

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